

Airframe Technology Master

Checkstress

Lecture 13. Metallic Sizing-Exercise (Distribution of loads)

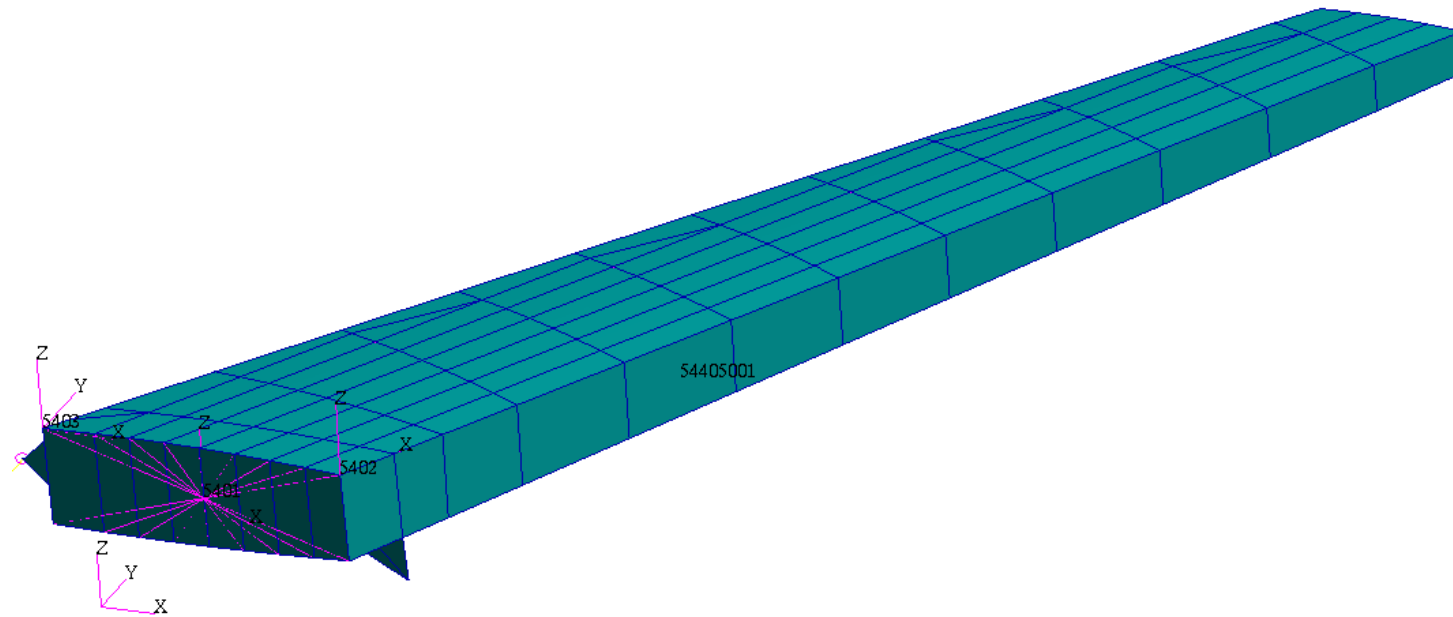
Andrés J. Ortiz Soriano

11 June 2025

Skin to Spar attachment (Exercise 1)

LOAD DISTRIBUTION Example

Load between Rear Spar and Upper Skin at Bay 4 (Element ID 54405001)

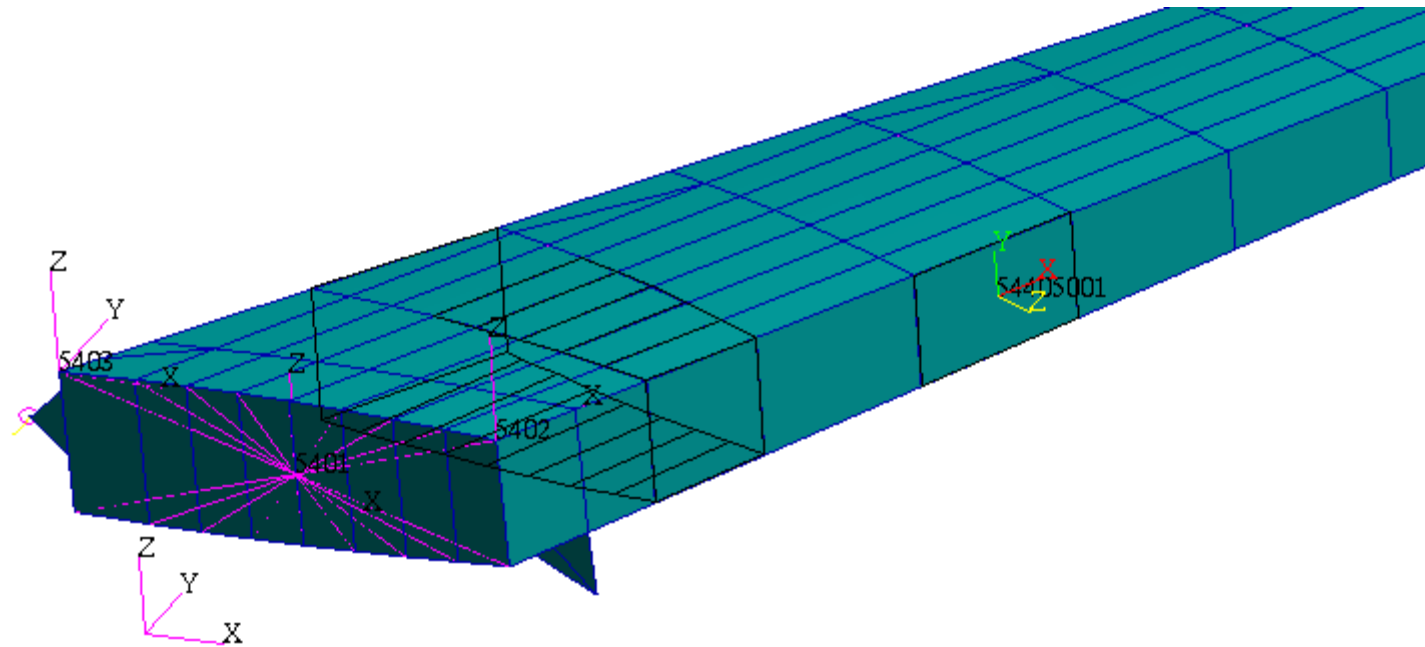


Distance between Ribs: 516 mm

LOAD DISTRIBUTION Example

Load between Rear Spar and Upper Skin at Bay 4 (Element ID 54405001)

First check: Element coordinate system



Distance between Ribs: 516 mm

LOAD DISTRIBUTION Example

From *.f06

```
FORCES IN QUADRILATERAL ELEMENTS (QUAD4)

ELEMENT      - MEMBRANE FORCES -      - BENDING MOMENTS -      - TRANSVERSE SHEAR FORCES -
  ID          FX          FY          FXY          MX          MY          MXY          QX          QY
54405001      5.434331E-01  2.490246E-01  6.871011E+00  0.0          0.0          0.0          0.0          0.0
54405001     -5.711386E-02 -7.904857E-02 -9.841164E+00  0.0          0.0          0.0          0.0          0.0
54405001      4.863192E-01  1.699760E-01 -2.970152E+00  0.0          0.0          0.0          0.0          0.0
54405001      3.689804E+00  1.585075E+00  2.841475E+01  0.0          0.0          0.0          0.0          0.0
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```

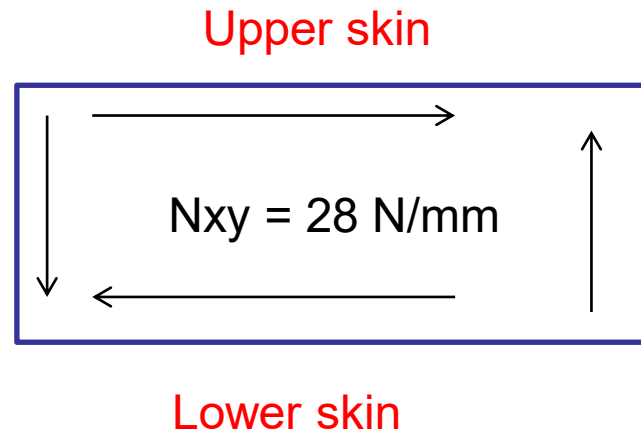
LC ID	Limit Load		
	Nx (N/mm)	Ny (N/mm)	Nxy (N/mm)
1	1	0	7
2	0	0	-10
3	0	0	-3
10	2	1	19

LC ID	Ultimate Load		
	Nx (N/mm)	Ny (N/mm)	Nxy (N/mm)
1	1	0	10
2	0	0	-15
3	1	0	-4
10	4	2	28

Distance between Ribs: 516 mm

LOAD DISTRIBUTION Example

Spar transfers the shear flux with the skin, so:



Total transfer load:

$N_{xy} \cdot \text{distance between ribs}$

$$P = 28 \cdot 516 = 14448 \text{ N}$$

With a pitch of 5D and $D=4,8 \text{ mm}$

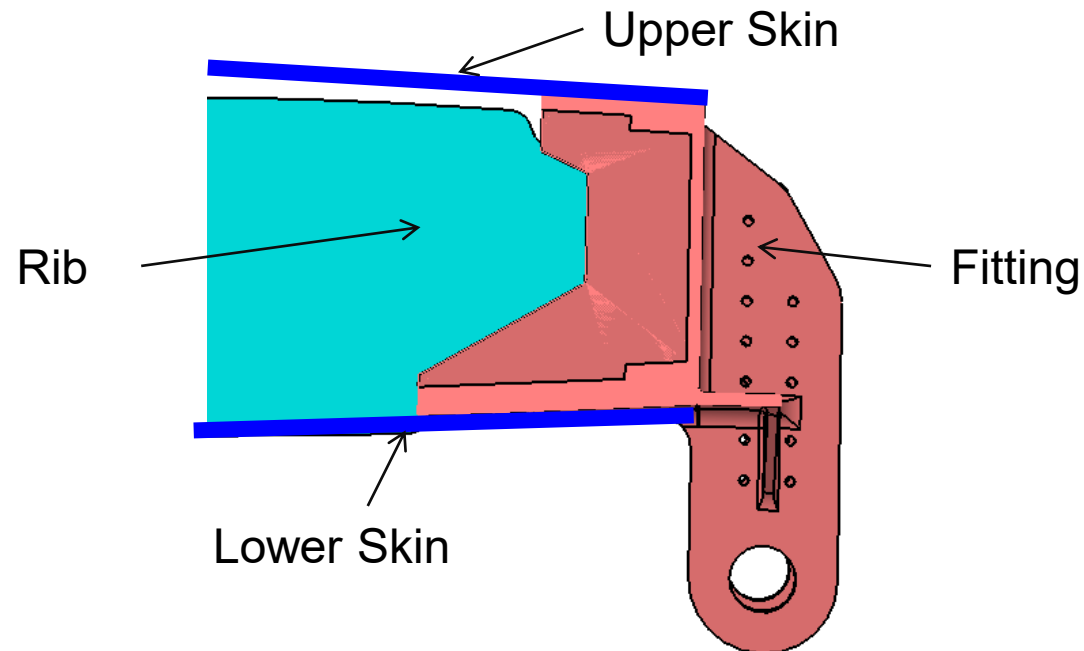
pitch = 24 mm

(2 rows of 21 bolts between ribs)

Load per bolt: $N_{xy} \cdot \text{pitch}$

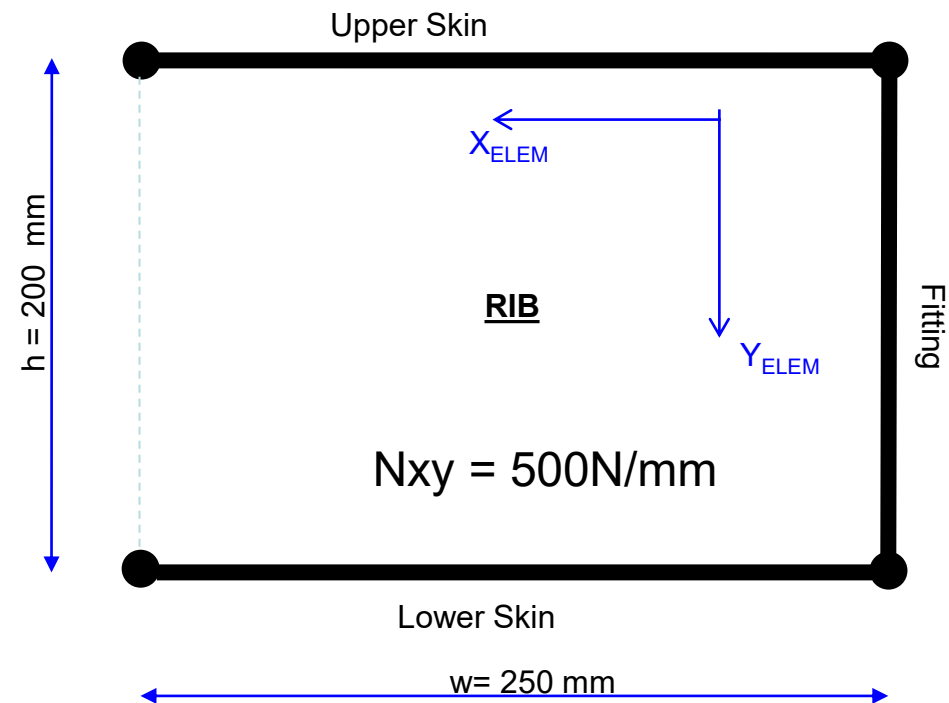
$$P_{sh} = 28 \cdot 24 / 2 = 336 \text{ N}$$

Fitting to Rib attachment (Exercise 2)



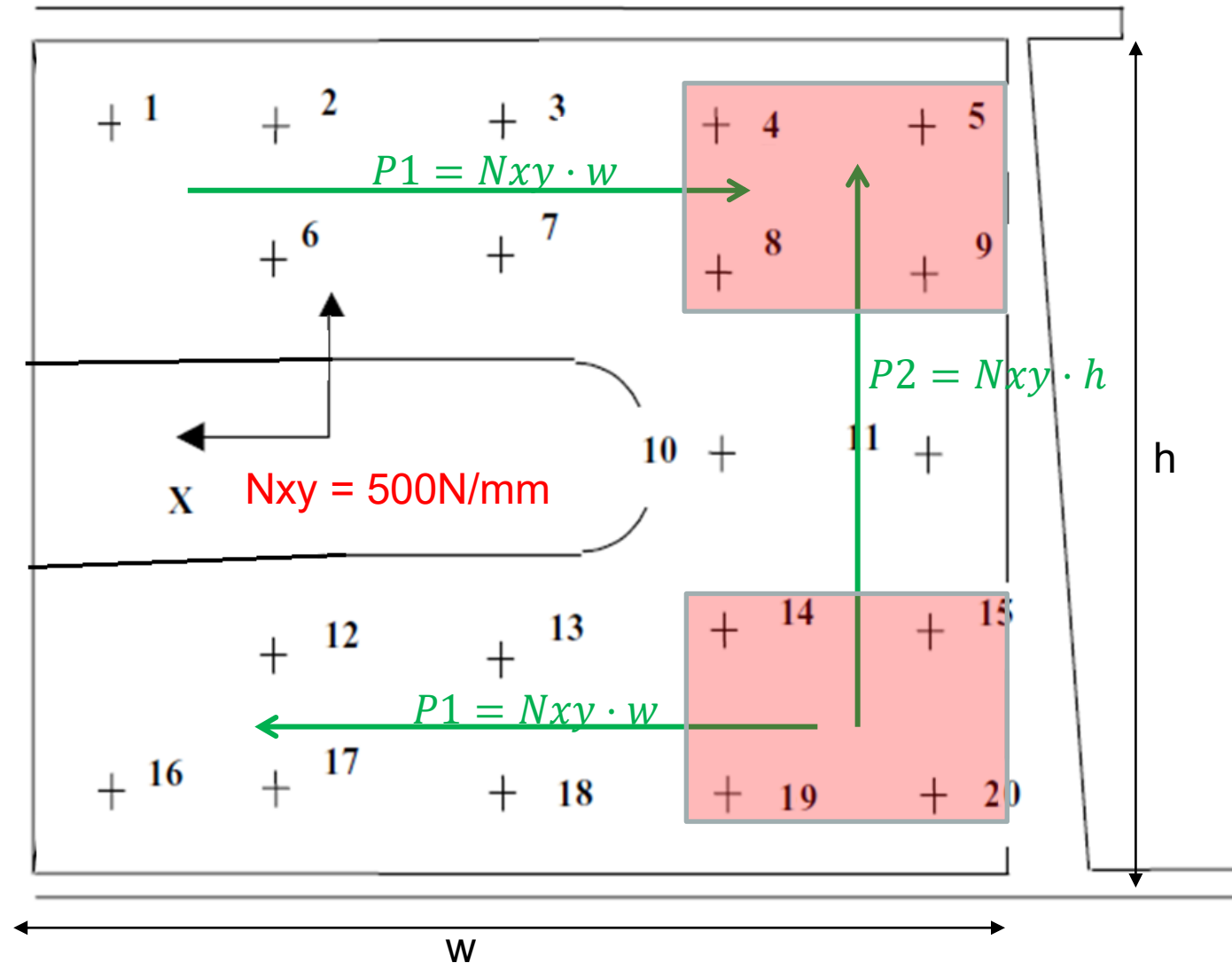
LOAD DISTRIBUTION Example

Load transfered by the Fitting to the Rib



LOAD DISTRIBUTION Example

Load transfered by the Fitting to the Rib



LOAD DISTRIBUTION Example

Load transfered by the Fitting to the Rib

$$P_{5,x} = \frac{P1}{n^{\circ}bolts} = \frac{500 \cdot 250}{9} = 13889N$$

$$P_{5,y} = \frac{P2}{n^{\circ}bolts} = \frac{500 \cdot 200}{10} = 10000N$$

$$P_{5,sh} = \sqrt{P_{5,x}^2 + P_{5,y}^2} = 17114N$$